MISSING *Female* ATHLETE DATA



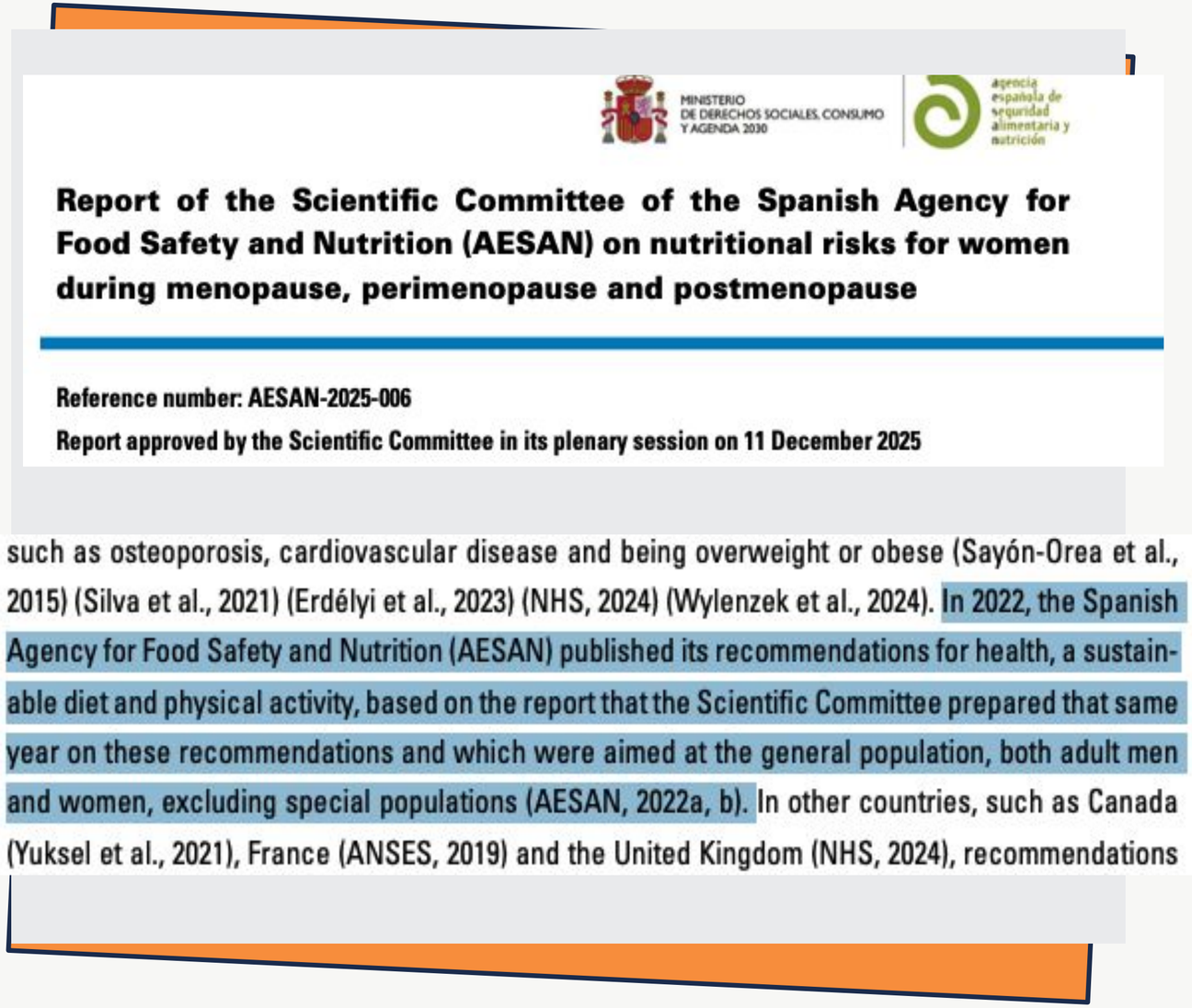
Swiss Python Summit

Hilal Işık

What happens when our models don't account for female physiology?

A REPORT ABOUT THE ONES *who go through*

PERIMENOPAUSE MENOPAUSE POSTMENOPAUSE



MINISTERIO DE DERECHOS SOCIALES, CONSUMO Y AGENDA 2030

agencia española de seguridad alimentaria y nutrición

Report of the Scientific Committee of the Spanish Agency for Food Safety and Nutrition (AESAN) on nutritional risks for women during menopause, perimenopause and postmenopause

Reference number: AESAN-2025-006

Report approved by the Scientific Committee in its plenary session on 11 December 2025

such as osteoporosis, cardiovascular disease and being overweight or obese (Sayón-Orea et al., 2015) (Silva et al., 2021) (Erdélyi et al., 2023) (NHS, 2024) (Wylenzek et al., 2024). In 2022, the Spanish Agency for Food Safety and Nutrition (AESAN) published its recommendations for health, a sustainable diet and physical activity, based on the report that the Scientific Committee prepared that same year on these recommendations and which were aimed at the general population, both adult men and women, excluding special populations (AESAN, 2022a, b). In other countries, such as Canada (Yuksel et al., 2021), France (ANSES, 2019) and the United Kingdom (NHS, 2024), recommendations

THE GENERAL RECOMMENDATION WASN'T THE END OF THE *Story*

2022 → 2025

AESAN recommendations for the general adult population

A specific assessment of nutritional risks during menopause

“...in order to ... establish a series of dietary and physical activity recommendations for women during these stages of life.”

- Insufficient intake of some nutrients, below the recommendations, is common in women in peri-menopause and post-menopause, especially in the case of vitamin D, calcium, magnesium and vitamins A, C, E and B. Deficiency of these nutrients contributes to the risk of osteoporosis, sarcopenia and increased morbidity.

Table 4. Adequate Intake (AI) and Nutrient Reference Intake Values (NRI), of nutrients, for the adult female population (≥40 years)

	AI	NRI	Reference
Minerals			
Calcium	-	950 mg/day	(EFSA, 2015a)
Iron	-	11 mg/day	(EFSA, 2015b)
Magnesium	300 mg/day	-	(EFSA, 2015c)
Vitamins			
Vitamin A	-	650 µg ER/day [1]	(EFSA, 2015d)
Vitamin B6	-	1.6 mg/day [2]	(EFSA, 2016a)
Vitamin B12	4 µg/day	-	(EFSA, 2015e)
Vitamin C	-	95 mg/day	(EFSA, 2013)
Vitamin D	15 µg/day [3]	-	(EFSA, 2016b)
Vitamin E	11 mg/day [4]	-	(EFSA, 2015f)

AESAN 2025 · TABLE 4

VITAMIN C

95

mg/ day

REFERENCE: EFSA, 2013

EFSA, 2013

and assuming a CV of 10 %, a PRI of 110 mg/day of vitamin C is derived for men. This intake is associated with a plasma ascorbate concentration above 50 µmol/L in almost all subjects of the population of healthy men.

Women reach a plateau for plasma concentrations at lower intakes than men. As no values for metabolic losses are available in women, the AR for women is extrapolated from the AR for men. This is done by isometric scaling (linear with body weight), since vitamin C is considered to be distributed throughout the whole body, since the multi-compartment models used to calculate the metabolic losses in men consider an exchange with only one whole body tissue pool, since few sex-related differences could be observed in the pharmacokinetics of vitamin C, and since a main part of the observed differences can be explained by body weight differences between sexes:

$AR_{women} = AR_{men}$ (weight of women/weight of men).

NO FEMALE METABOLIC-LOSS DATA

extrapolate from men

MALE DATA → MODEL → FEMALE NUMBER

WHOSE BODY ARE WE DESIGNING FOR?

Running shoes

SO WHAT WAS THE SHOE BASED ON?

424 female feet were compared with 4 running-shoe lasts



“women's lasts are often based on men's lasts, scaled down to fit women's feet”

The study found substantial differences in several width and height measures between the lasts and female foot types.

Krauss et al. (2010). *Research in Sports Medicine, 18*(2), 140–156.

Where are all the female participants in Sports and Exercise Medicine research?

Joseph T Costello [1], Francois Bieuzen, Chris M Bleakley

Affiliations + expand

PMID: 24766579 DOI: 10.1080/17461391.2014.911354

Free article

Abstract

The aim of this study is to estimate the ratio of male and female participants in Sports and Exercise Medicine research. Original research articles published in three major Sports and Exercise Medicine journals (Medicine and Science in Sports and Exercise, British Journal of Sports Medicine and American Journal of Sports Medicine) over a three-year period were examined. Each article was screened to determine the following: total number of participants, the number of female

WHERE ARE ALL THE *female* PARTICIPANTS?

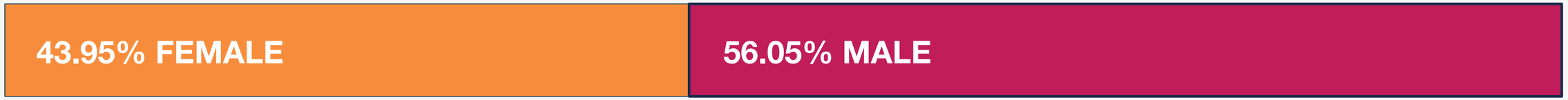
1,382
ARTICLES

6.08 million
PARTICIPANTS

COSTELLO, BIEUZEN & BLEAKLEY, 2014



A DECADE *later...*



5.6%

ONLY 5.6% OF STUDIES CONSIDERED MENSTRUAL STATUS IN THEIR STUDY DESIGN.

Where Are All the Female Participants in Sports and Exercise Medicine Research? A Decade Later

Benjamin M Ose [1], Jake Eisenhauer [1], Isaiah Roepe [2], Ashley A Herda [2 3], Bryan G Vopat [2], Lisa M Vopat [2]

Affiliations + expand

PMID: 39780766 DOI: 10.1177/03635465241278350

Abstract

Background: In 2014, Costello and colleagues published a sentinel paper spotlighting the large disparity of female versus male representation within sports science and sports medicine (SSSM) research.

Purpose: To (1) revisit the method published by Costello et al a decade later to evaluate female representation and (2) ascertain whether study designs account for menstrual status.

Study design: Systematic Review; Level of evidence, 4

MEIGNIÉ, TOUSSAINT & ANTERO, 2022

MENSTRUAL CYCLE AS "THE PROBLEM"

FEMALE PHYSIOLOGY CAN BE TREATED AS A CONFOUNDING VARIABLE. JUST LIKE TEMPERATURE, NUTRITION, OR FATIGUE.

Dealing with menstrual cycle in sport: stop finding excuses to exclude women from research

Alice Meignié [1], Jean-François Toussaint [2 3], Juliana Antero [2]

Affiliations + expand

PMID: 35984494 DOI: 10.1007/s00421-022-05014-1

Abstract

The current literature is mostly male-based, limiting evidence-based recommendations for training individualization for female athletes. Recently, studies have relied on recent findings showing a potential effect of the menstrual cycle to exclude female athletes from their samples. We highlight that the arguments usually put forward to this action are not acceptable. Our discussion aims to

HEALTHY YOUNG ADULTS:

24 physically active male subjects.



Home > European Journal of Applied Physiology > Article

Fasting-Mimicking-Diet does not reduce skeletal muscle function in healthy young adults: a randomized control trial

Original Article | Published: 16 January 2022

Volume 122, pages 651–661 (2022) Cite this article

Access provided by Universidade de Aveiro, member of B-on Consortium Portugal

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Mauro Nardon, Massimo Venturelli, Federico Ruzzante, Valter D. Longo & Matteo Bertucco

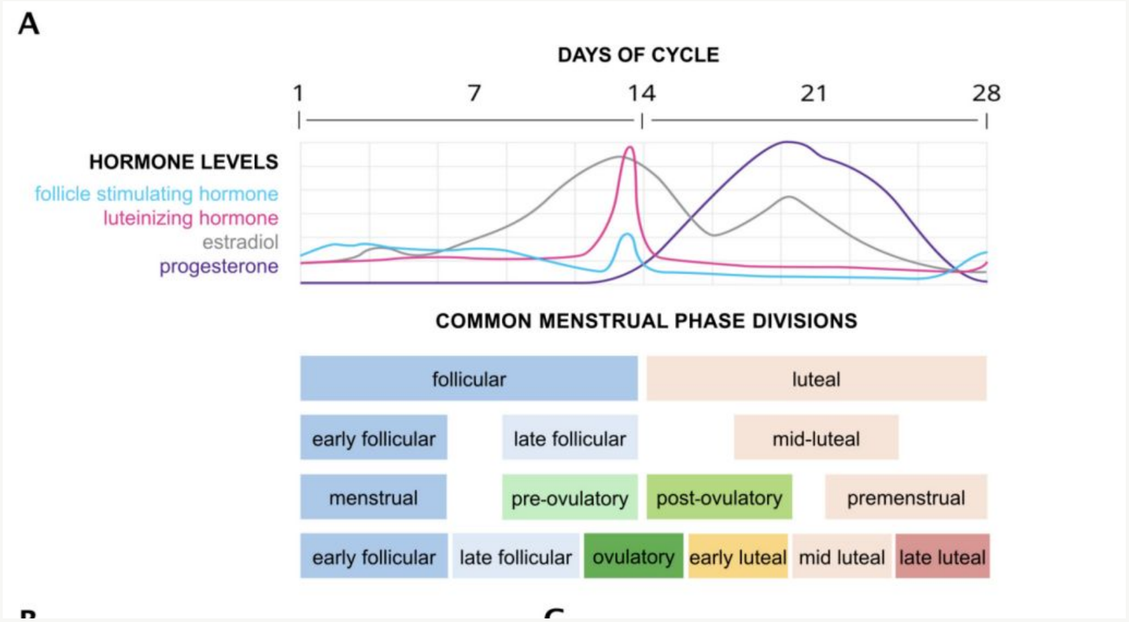
"We explicitly excluded female subjects from the study to avoid the possible influence of the menstrual cycle, since the menstrual cycle hormonal fluctuation could be a confounder on both metabolic patterns influencing body composition *(Davidsen et al. 2007; Wallace et al. 2010; Draper et al. 2018) and exercise performance (McNulty et al. 2020; Meignié et al. 2021). Future studies with larger sample size should address the effects of FMD also in this sex subgroup, possibly controlling testing by menstrual cycle phases."*

MODELLING *the cycle*

1- PHASE CATEGORIES
`C(phase)`

2- SINE / COSINE
`sin(2πt/L) + cos(2πt/L)`

3- GAM
`s(cycle_day, basis="cp")`



WHAT HAPPENS IF I MODEL *one woman* PROPERLY?

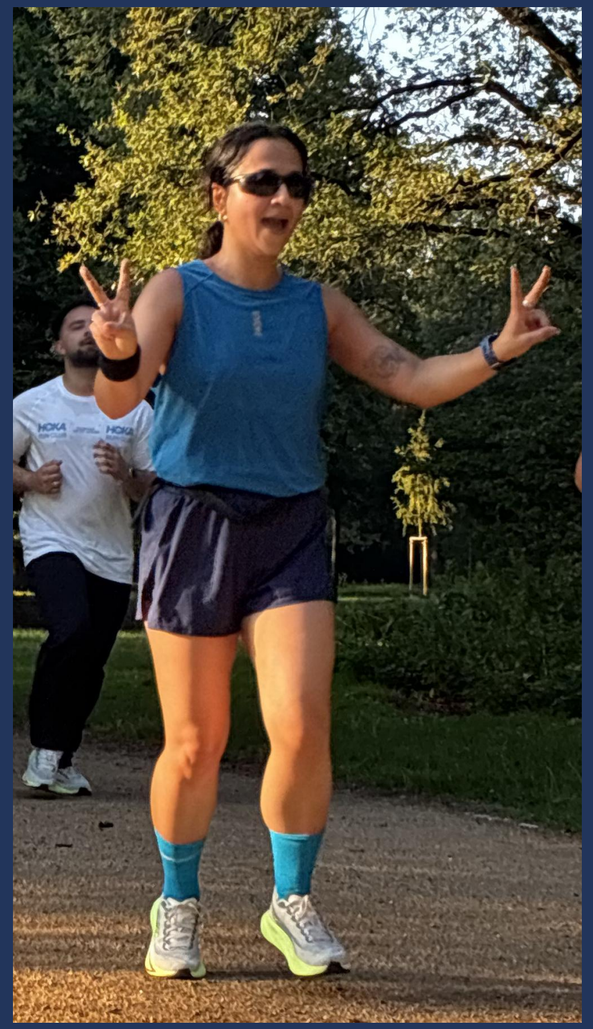


Photo credit: Brice Maillard, Hoka Running club 2026

We do *sports(!)*

- **England: women's football ban 1921–1970/71**
  The FA barred women from playing on affiliated club grounds after women's matches had attracted huge crowds. The FA said football was "quite unsuitable for females."

- **West Germany: women's football ban 1955–1970**
  The German Football Association banned women's football, citing health, fertility and propriety. The ban was lifted in 1970.

- **The US: Boston Marathon Women banned from officially competing until 1972**
  The Amateur Athletic Union (AAU) prohibited women from competing in sanctioned races longer than 1.5 miles. The justification was that women were not physiologically capable of running a marathon, with concerns that long-distance running could harm their reproductive health

https://www.thefa.com/womens-girls-football/heritage/kicking-down-barriers
https://nationalfootballmuseum.com/items/the-fa-minute-book-1921/

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THANK YOU

Let's Connect

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Swiss Python Summit

Hilal Işık